

AERO 4510: Propulsion Fall 2025

Fundamentals of Rocket Propulsion

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Apollo 11

NASA Image

Rocket Fundamentals

(Chemical and Thermal rockets)

- **Nozzle Flow**
- **Thrust Equation**
- **Effective Exhaust Velocity**
- **Total Impulse**
- **Specific Impulse**
- **Mass Ratio**
- **Characteristic Velocity**
- **Thrust Coefficient**
- **Summary of Performance**
- **Standard Atmosphere**
- **The Rocket Equation**
- **Staging**

Comments Regarding Units

- Many government agencies have adopted the SI system universally; however, some government agencies including NASA's engineering design teams and many industrial units still use the English system either exclusively or extensively. Therefore this short course will include examples based on both sets of units.
- Two key points to remember which will likely help you to sort out units issues with the English system are:

$$1 \text{ slug} = 32.17 \text{ lbm}$$

$$1 \text{ lb}_f = \frac{\text{slug ft}}{\text{sec}^2}$$

The Thrusting Nozzle: Unknowns and Physical Principles

- **Consider gas flowing through a Laval Nozzle from a known stagnation state to produce thrust:**
- **What are the unknowns?**
 - **Thrust**
 - **Thermodynamic state at the nozzle exit**
 - **Pressure**
 - **Temperature**
 - **Density**
 - **Bulk Flow Velocity**
- **A total of 5 unknowns for the steady flow case.**

The Thrusting Nozzle: Unknowns and Physical Principles

- **What physical principles can be used to develop 5 equations which can be used to find these 5 unknowns?**
 - **Conservation Equations**
 - **Continuity**
 - **Momentum (ultimately used to determine thrust)**
 - **Energy (1st Law)**
 - **Equation of State**
 - **Second Law of Thermodynamics**
- **A total of 5 physical principles for the steady flow case.**

Nozzle Flow Performance Review: Isentropic Flow of a Perfect Gas

- Definition of Entropy

$$dS = \left[\frac{dQ}{T} \right]_{rev}$$

- For Irreversible Processes

$$dS = dS_{irrev} + \frac{dQ}{T}$$

- dS_{irrev} is always positive while dQ/T has the sign of dQ so

$$dS \geq \frac{dQ}{T}$$

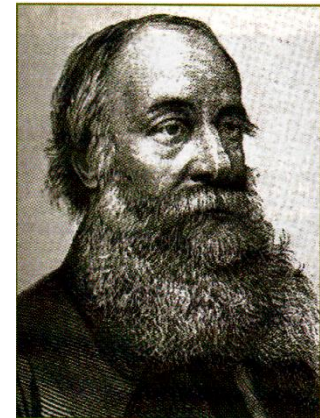
If the process is reversible and adiabatic, it is isentropic

First Law of Thermodynamics

$$dq = de + dw$$

where q is heat into a system per unit mass, e is internal energy due to random thermal motion per unit mass and w is work out per unit mass

Historical Note: James Joule established the equivalence of heat and work in the 19th century. His tombstone is inscribed with a number 722.55, which approximates the now accepted value of 778.16 ft-lb_f/Btu.



Source: Mechanical Engineering magazine, July 2007.

Gibbs Equation

(Fundamentally, an extension of the second law of thermodynamics)

$$Tds = de + pdv$$

This equation holds for “pure” substances (substances with 2 independent state variables) Note that

$$h = e + pv$$

$$dh = de + pdv + vdp$$

So the Gibbs Equation becomes

$$Tds = dh - vdp$$

Historical Note: Second Law Development

- **The classical treatment of the second law of thermodynamics was developed in phases:**
 - **Phase I – Sadi Carnot (1796 - 1832) established the principle of ideal limiting processes which occur when irreversibilities approach zero and the principle of combining a sequence of such processes such that the endpoint of the last was again at the beginning of the first. The thermodynamic cycle was born.**
 - **Phase II – William Thompson (1821-1907) (later, Lord Kelvin) established the absolute temperature scale.**
 - **Phase III – Rudolph Clausius from Germany modified Kelvin's work and established the concept of positive heat in and positive work out.**

Two additional phases referring to substances and chemical reactions occurred in the 20th century.

Equation of State

- For a Perfect Gas, the equation of state can be written in a variety of ways including:

$$p = \rho RT$$

$$pv = RT = \frac{R_u}{MW} T$$

where

$$R_u = 1545 \frac{ft \ lb_f}{lb_m - mole} \quad R = 8314 \frac{J}{kg - mole \ K}$$

Definition of Specific Heats

$$c_v = \frac{de}{dT} \quad c_p = \frac{dh}{dT}$$

From the definition of enthalpy

$$dh = de + RdT$$

Substituting

$$c_p = c_v + R$$

Calorically or Thermally Perfect Gases

- **Calorically Perfect Gases**

$$\frac{de}{dT} = \text{const.}, \quad \frac{dh}{dT} = \text{const.} \quad \text{and} \quad h = e + RT$$

- **Thermally Perfect Gases**

$$c_v = c_v(T), \quad c_p = c_p(T) \quad \text{and} \quad h = e + RT$$

Isentropic Processes

From the Gibbs Equation,

$$Tds = dh - vdp = dh - \frac{dp}{\rho}$$

Or

$$ds = c_p \frac{dT}{T} - R \frac{dp}{p}$$

For an isentropic process

$$\frac{p}{p_1} = \left(\frac{T}{T_1} \right)^{c_p/R}$$

Specific Heat Ratio

$$\gamma = c_p / c_v \text{ and } c_p / R = c_p / (c_p - c_v) = \gamma / (\gamma - 1)$$

for an isentropic process

$$\frac{p}{p_1} = \left(\frac{T}{T_1} \right)^{\gamma / (\gamma - 1)} = \left(\frac{\rho}{\rho_1} \right)^{\gamma}$$

Nozzle Flow

- Equations developed here relate local thermodynamic state variables to stagnation states
- Mach Number and Sound Speed

$$M = \frac{\text{local velocity}}{\text{local sound speed}} = \frac{u}{a}$$

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho} \right)_s} = \sqrt{\gamma RT}$$

Note: Ernst Mach (1838-1916) conducted a series of experiments during the period 1873-1893 at the University of Prague in the area of supersonic flow over projectiles and the propagation of sound waves. He is responsible for the development of the schlieren and several other interferometric techniques. He was the first to photograph a shock wave at essentially the Mach angle and to demonstrate the discontinuities associated with waves traveling beyond $v/a = 1$.

Isentropic Flow

For adiabatic flow with no work

$$h_o = h + \frac{u^2}{2} \Rightarrow c_p T_o = c_p T + \frac{u^2}{2}$$

Rearranging and using the definition of Mach Number

$$\frac{T_o}{T} = 1 + \frac{u^2}{2c_p T} = 1 + \frac{\gamma - 1}{2} M^2$$

This holds for all adiabatic zero-work processes whether they are isentropic or not

For flows which are isentropic, the isentropic relationship between temperature pressure and density can be used to produce

$$\frac{p_o}{p} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

$$\frac{\rho_o}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{1}{\gamma - 1}}$$

These three equations relate the local static properties to the local stagnation properties regardless of whether the flow is isentropic or not.

In order to relate the thermodynamic properties to the nozzle dimensions, an area-Mach number relationship is needed. Using continuity, we can write

Where

$$\frac{\dot{m}}{A} = \rho u = \rho_o \frac{\rho}{\rho_o} u$$

$$u = aM = M \sqrt{\gamma R T_o \left(\frac{T}{T_o} \right)} = M \sqrt{\frac{\gamma R T_o}{1 + \frac{\gamma - 1}{2} M^2}}$$

Using the isentropic relationship between density and Mach number

$$\frac{\dot{m}}{A} = \frac{p_o \sqrt{\gamma}}{\sqrt{RT_o}} M \left(\frac{1}{1 + \frac{\gamma-1}{2} M^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

This is an Area-Mach number relationship containing only constants. The mass flow rate can be eliminated by using as a reference condition the minimum value for area (throat where $M = 1$).

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Maximum Flow Velocity

- The maximum achievable velocity occurs when the local static temperature and corresponding random motion go to negligibly small values.

$$h_o = h + \frac{u^2}{2} \Rightarrow c_p T_o = c_p T + \frac{u^2}{2}$$

- Hence, the maximum achievable velocity is

$$u_{\max} = \sqrt{2c_p T_o} = \sqrt{\frac{2\gamma R T_o}{\gamma - 1}}$$

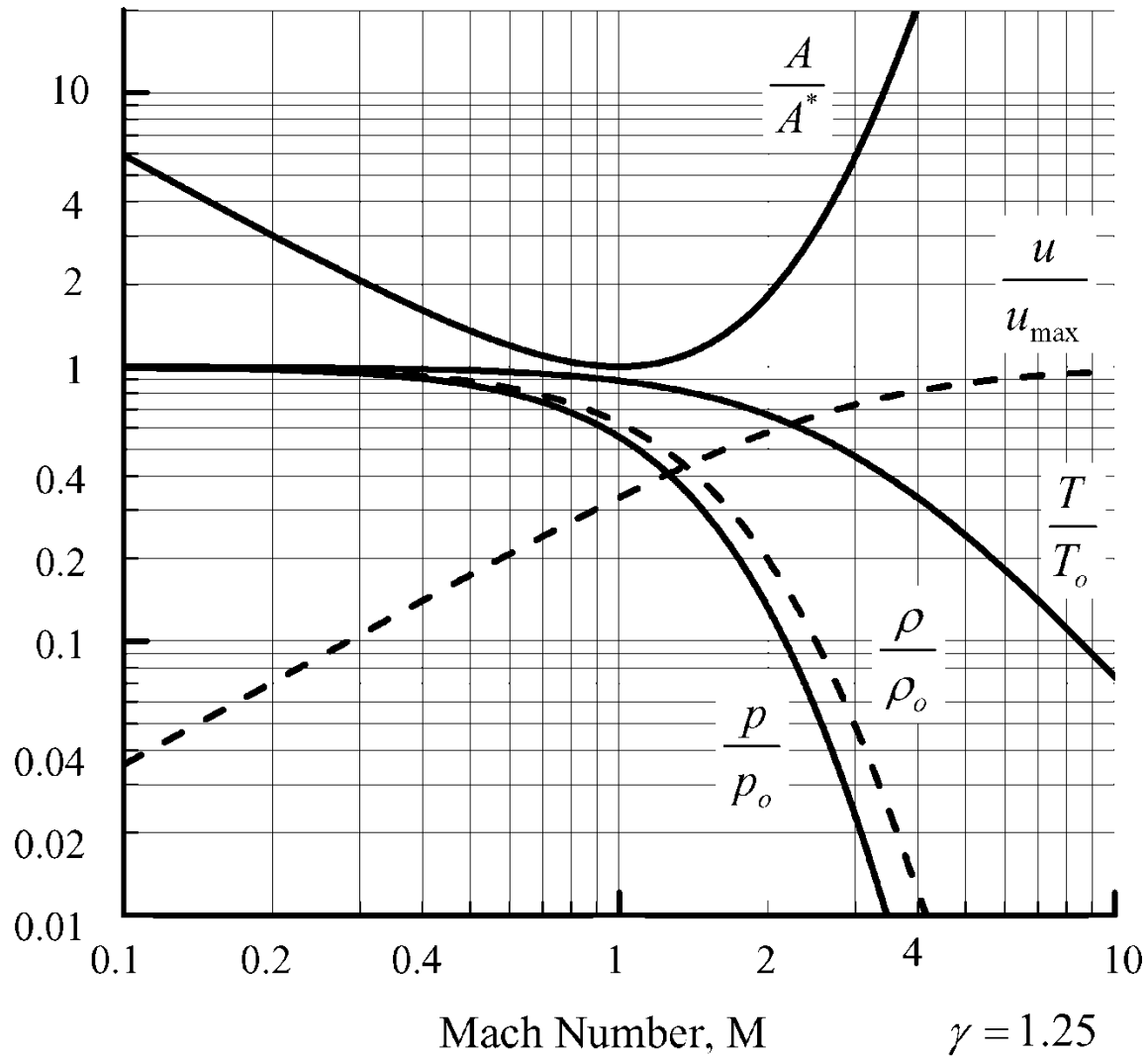
Normalizing by $u_{\max}$

$$\frac{u}{u_{\max}} = M \sqrt{\frac{\gamma - 1}{2} \left(\frac{1}{1 + \frac{\gamma - 1}{2} M^2} \right)}$$

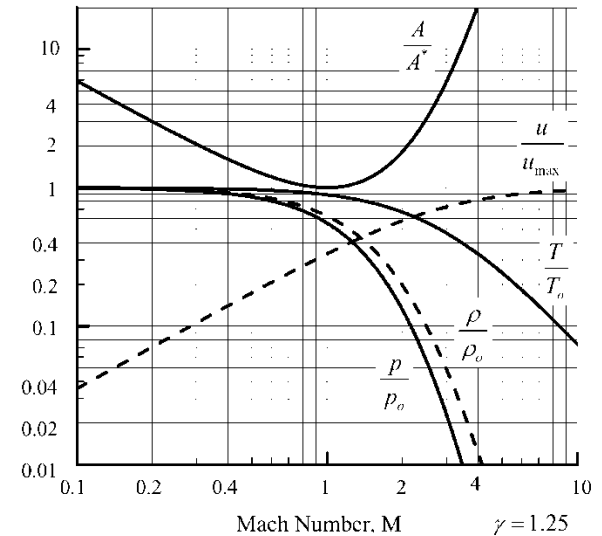
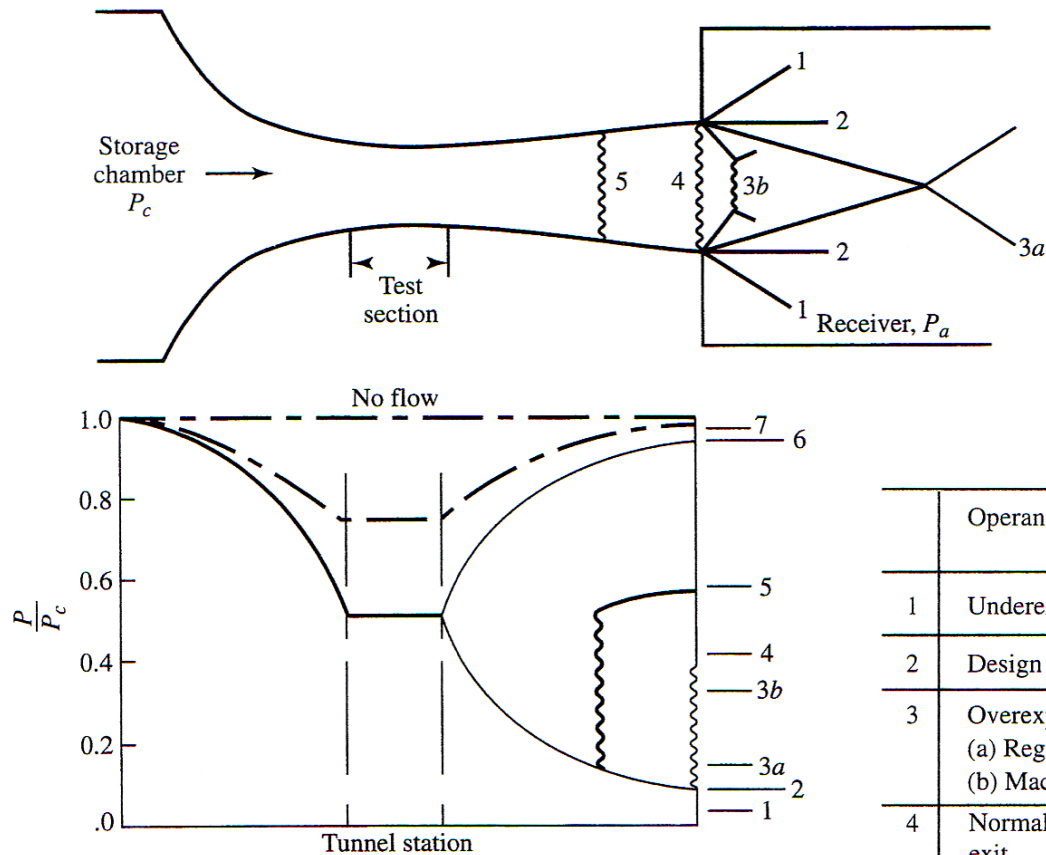
Footnote: Isentropic Chain

$$\frac{T_o}{T} = \left(\frac{p_o}{p} \right)^{\frac{\gamma - 1}{\gamma}} = \left(\frac{\rho_o}{\rho} \right)^{\gamma - 1} = \left(\frac{A}{A^*} M \right)^{\frac{2(\gamma - 1)}{\gamma + 1}} \frac{\gamma + 1}{2} = \frac{2}{\gamma - 1} \left(M \left(\frac{u_{\max}}{u} \right) \right)^2$$

Graphical Display of Isentropic Relations



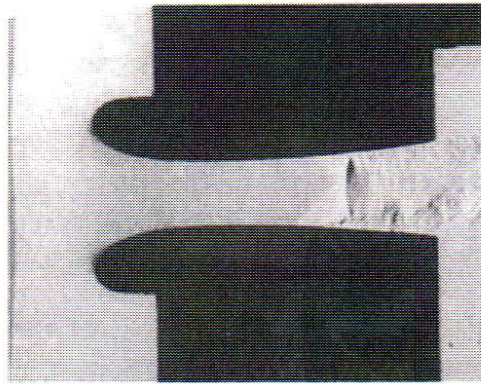
Converging-Diverging Nozzle Pressure Distribution



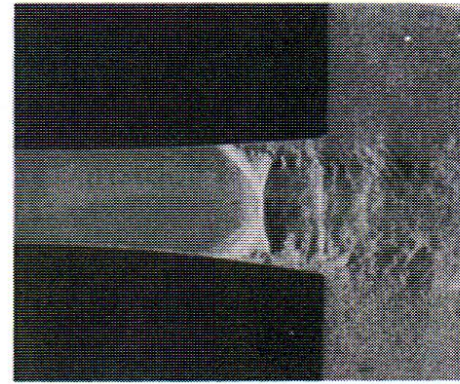
	Operand point	Exit section pressure P_e	Nozzle pressure ratio $P_n = P_c/P_a$	Mass flow rate	
1	Underexpanded	$P_e > P_a$	$P_e = \frac{P_c}{P_{\bar{n}}}$	$P_n > P_{\bar{n}}$	Maximum
2	Design	$P_e = P_a$		$P_n = P_{\bar{n}}$	Maximum
3	Overexpanded (a) Reg. reflection (b) Mach reflection	$P_e < P_a$		$P_n < P_{\bar{n}}$	Maximum
4	Normal shock at exit	$(P_e)_x < (P_e)_y = P_a$		$P_n < P_{\bar{n}}$	Maximum
5	Normal shock in divergent section	$P_e = P_a$		$P_n > P_{\bar{n}}$	Maximum
6	Sonic at throat subsonic elsewhere	$P_e = P_a$		$P_n < P_{\bar{n}}$	Maximum
7	Subsonic everywhere	$P_e = P_a$		$P_n < P_{\bar{n}}$	Less than Maximum

Mattingly, p. 170

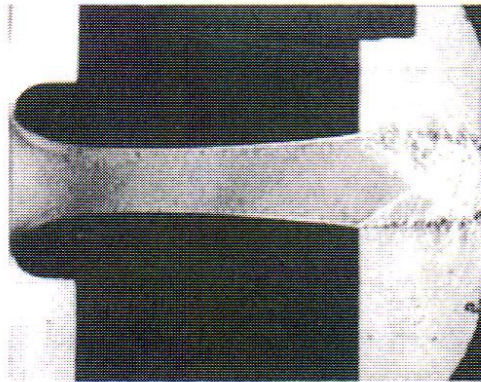
Converging-Diverging Nozzle Pressure Distribution



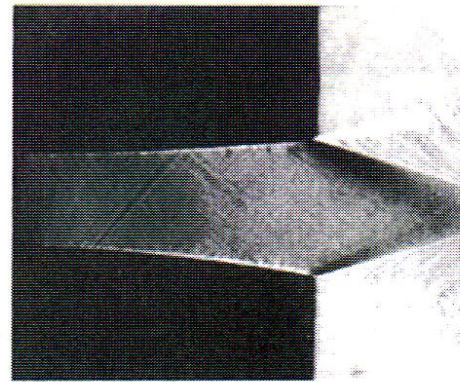
(a) $P_n = 1.5$, normal shock, inside



(b) $P_n = 2.5$, Mach reflection, overexpansion



(c) $P_n = 4.5$, regular reflection, overexpansion

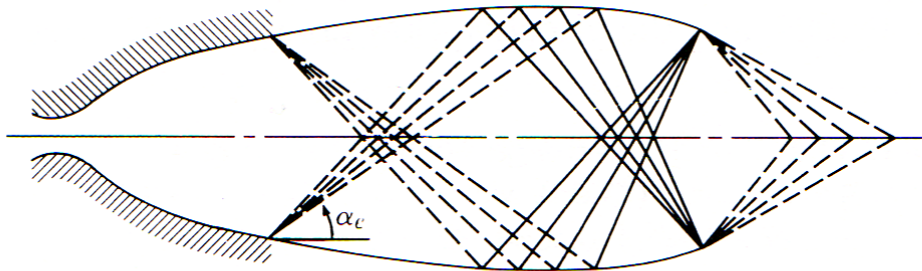


(d) $P_n = 8.0$, underexpansion

FIGURE 3-47

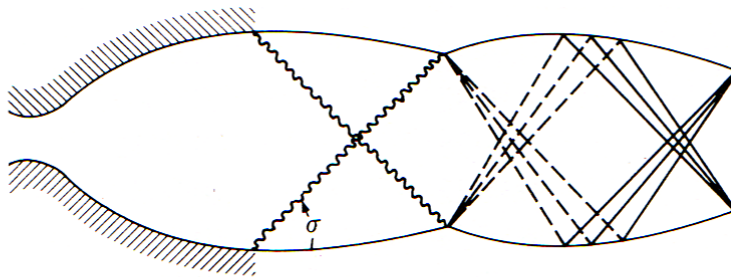
Spark Schlieren photographs of nozzle exhaust flow patterns for an area ratio of 1.5. (*Department of Aero-Mechanical Engineering, AFIT, WPAFB, Ohio.*)

Under- Over- Expansion of Nozzles



(a) Underexpanded flow

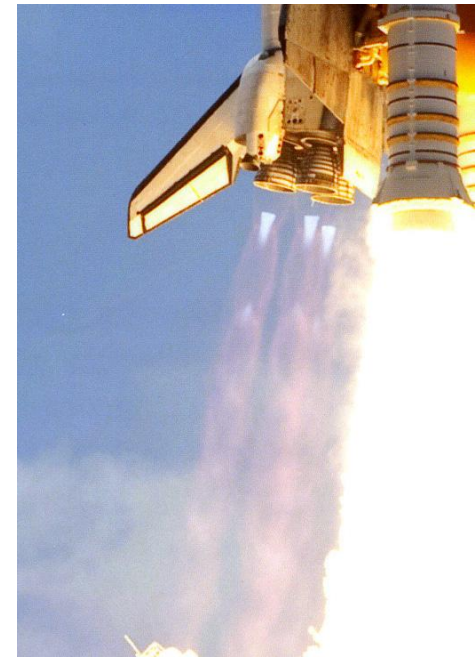
--- Expansion waves
— Compression waves
~~~~~ Shock waves



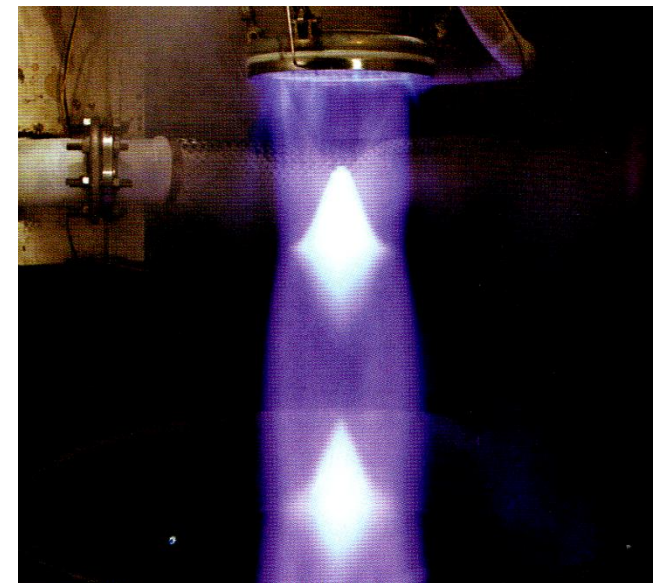
(b) Slightly overexpanded flow

**FIGURE 11.17** Exhaust jet behavior at various back pressures.

Hill & Peterson p. 536

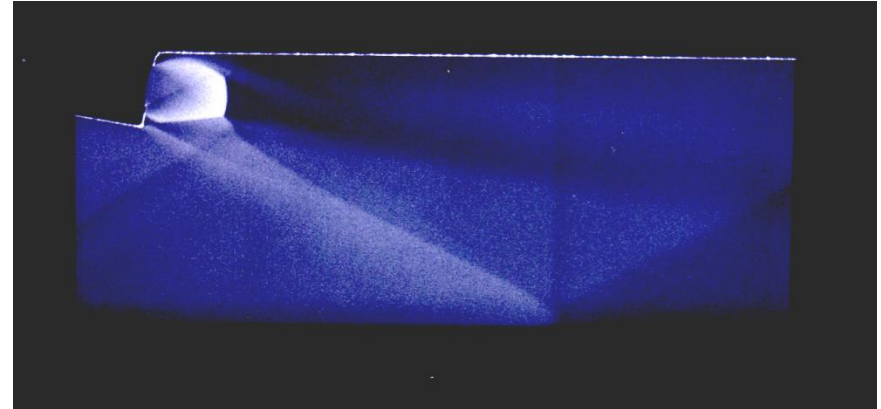


**Atlantis,  
STS 132**



**Photo of  
overexpanded  
nozzle test**

# Under- Expanded Jet Images



Hartfield, R. J., Jr., "Planar Measurement of Flow Field Parameters in Nonreacting Supersonic Flows with Laser-Induced Iodine Fluorescence," PhD. Dissertation, Department of Mechanical and Aerospace Engineering, University of Virginia, May 1991.

## Performance Example 1 (Ideal Nozzle)

**Consider a rocket motor equipped with a nozzle in which the stagnation conditions are 3000 K and 30 atm. The specific heat ratio is 1.25 and the molecular weight of the propellant is 18. The nozzle throat diameter is 10 cm and the nozzle exit diameter is 15 cm. Calculate the pressure and velocity at the nozzle exit.**

## **Solution:**

### **Step 1: Find Mach number from A-M relationship**

$$\frac{1}{M_e} \left[ \frac{2}{\gamma + 1} \left( 1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} = \frac{A_e}{A^*} = \frac{\pi D_e^2 / 4}{\pi D_t^2 / 4} = \left( D_e / D_t \right)^2$$

**This equation must be solved iteratively and has a subsonic and a supersonic solution. The supersonic solution applies if a sufficient pressure ratio exist. For the 2.25 area ratio, the supersonic solution for the Mach number is 2.20. Pressure can then be obtained from**

$$\frac{p_o}{p_e} = \left( 1 + \frac{\gamma - 1}{2} M_e^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

**For M = 2.20, the ratio is 10.65 and the exit pressure is 2.82 atm.  
For a 30 atm stagnation condition**

**To get velocity, temperature is required.**

$$\frac{T_o}{T_e} = 1 + \frac{\gamma - 1}{2} M_e^2$$

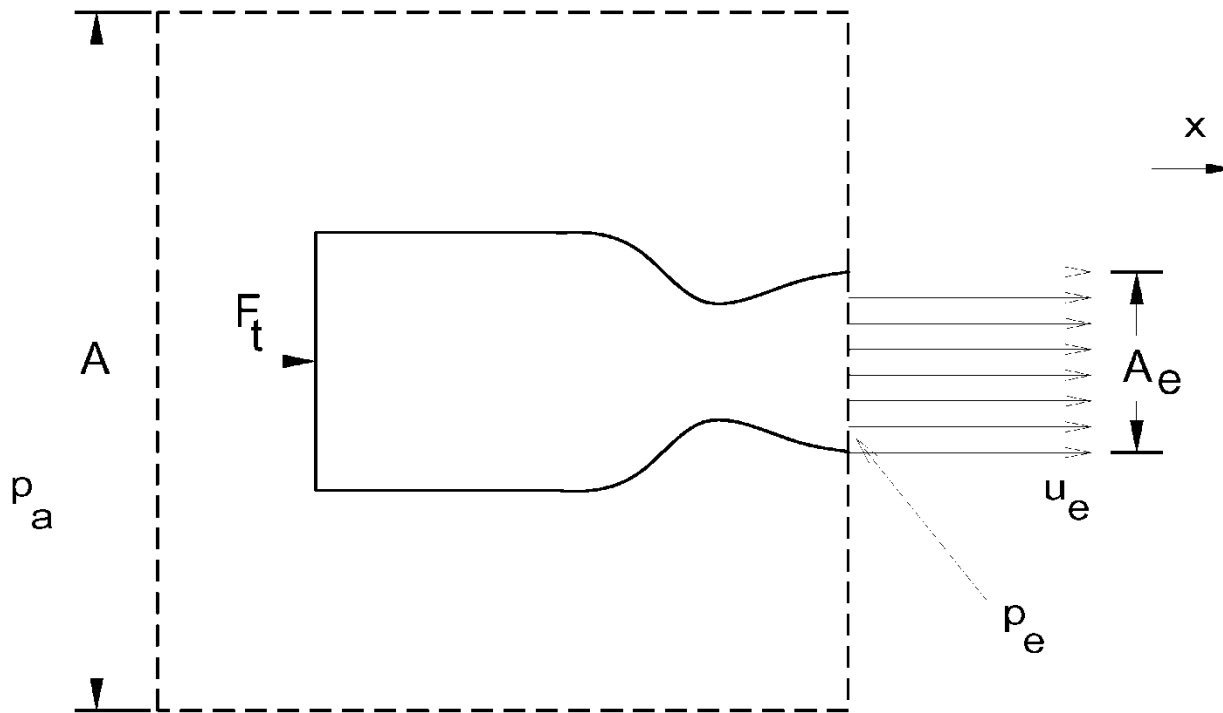
**For  $M = 2.20$ , this ratio is 1.605 and for a total temperature of 3000 K, the exit temperature is 1869 K. The exit velocity can then be calculated from the Mach number and local sound speed.**

$$u_e = Ma = M \sqrt{\frac{\gamma R_u}{MW} T} = 2.20 \sqrt{1.25 \frac{8314}{18} 1869} = 2285 \text{ m/s}$$



# Thrust Equation (for a Rocket)

- Apply the Momentum Equation to a Control Volume



## Thrust Equation (for a Rocket)

### Momentum Equation (Newton's 2<sup>nd</sup> Law for a Control Volume)

$$\frac{d}{dt} \int_{cv} \rho \vec{u} dV + \int_{cs} \vec{u} (\rho \vec{u} \cdot \hat{n}) dA = \sum \vec{F}$$

### Steady, Uniform 1-D Momentum Equation

$$\int_{cs} (\rho u) u dA = \dot{m} u|_{out} - \dot{m} u|_{in} = \sum F_x$$

$$\sum F_x = F_t + p_a A - p_a (A - A_e) - p_e (A_e) = F_t + A_e (p_a - p_e)$$

**For a steady stream at a given free stream velocity**

**Or**

$$\dot{m} u_e = F_t + A_e (p_a - p_e)$$

$$F_t = \dot{m} u_e + A_e (p_e - p_a)$$

## Performance Example 2 (Thrust)

Consider a rocket with a throat diameter of 1” and a nozzle design point at sea level. The propellant has burn temperature = 6059°R, molecular weight = 24.3, and the specific heat ratio = 1.24. The rocket is operated with a chamber pressure of 94.8 psia. Calculate the thrust assuming the rocket is fired at sea level.

### Solution:

From the problem statement, the exit pressure is 14.7 psia and there is no pressure term in the thrust equation so we can write:

$$T = \dot{m}u_e$$

# Performance Example 2 (Thrust) Solution

**The Nozzle Mass Flow rate can be written as:**

$$\frac{\dot{m}}{A} = \frac{p_o \sqrt{\gamma}}{\sqrt{RT_o}} M \left( \frac{1}{1 + \frac{\gamma-1}{2} M^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

$$\begin{aligned} \dot{m} &= A^* \frac{p_o \sqrt{\gamma}}{\sqrt{RT_o}} \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} = A^* \frac{p_o \sqrt{\gamma}}{\sqrt{\frac{R_u}{MW} T_o}} \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \\ &= \left( \frac{\pi (1 \text{ in})^2}{4} \right) \frac{\left( 94.8 \frac{\text{lb}_f}{\text{in}^2} \right) \sqrt{1.24}}{\sqrt{\frac{1545.43 \frac{\text{ft lb}_f}{\text{lb}_m - \text{mole}^\circ \text{R}}}{24.3 \frac{\text{lb}_m}{\text{lb}_m - \text{mole}}} (6059^\circ \text{R}) \left( 32.2 \frac{\text{lb}_m}{\text{slug}} \right)}} \left( \frac{2}{1.24+1} \right)^{\frac{1.24+1}{2(1.24-1)}} \left( 32.2 \frac{\text{lb}_m}{\text{slug}} \right) \\ &= 0.447 \frac{\text{lb}_m}{\text{sec}} \end{aligned}$$

## Performance Example 2 (Thrust) Solution

- The exit velocity is determined from the pressure ratio, the propellant characteristics, and the total Temperature.

$$T_{exit} = T_o \left( \frac{p}{p_o} \right)^{\frac{\gamma-1}{\gamma}} = 6059^\circ R \left( \frac{14.7 \text{ psia}}{94.8 \text{ psia}} \right)^{\frac{1.24-1}{1.24}} = 4224^\circ R$$

$$u_e = \sqrt{2(h_o - h_e)} = \sqrt{2c_p(T_o - T_{exit})} = \sqrt{2 \frac{\gamma}{\gamma-1} \frac{R_u}{MW} (T_o - T_{exit})}$$

$$u_e = \sqrt{(2) \left( \frac{1.24}{1.24-1} \right) \left( \frac{1545 \frac{\text{ft} \cdot \text{lb}_f}{\text{lb}_m - \text{mole}^\circ R}}{24.3 \frac{\text{lb}_m}{\text{lb}_m - \text{mole}}} \right) \left( 32.2 \frac{\text{lb}_m}{\text{slug}} \right) (6059^\circ R - 4224^\circ R)}$$
$$= 6232 \text{ ft/sec}$$

## Performance Example 2 (Thrust) Solution

**Thrust can now be calculated**

$$T = \dot{m}u_e = \left(0.447 \frac{lb_m}{\text{sec}}\right) \left(\frac{1}{32.2} \frac{\text{slug}}{lb_m}\right) \left(6232 \frac{ft}{\text{sec}}\right) = 86.51 \text{ } lb_f$$

# Effective Exhaust Velocity

- It can be readily shown that the maximum thrust occurs when the exit pressure matches the ambient pressure.
- For a rocket in which this is not the case, the thrust can be written as:

$$F_t = \dot{m} \left[ u_e + \frac{A_e}{\dot{m}} (p_e - p_a) \right] = \dot{m} c$$

Where the effective exhaust velocity is:

$$c = u_e + \frac{A_e}{\dot{m}} (p_e - p_a)$$

# Total Impulse

- The total impulse of a rocket is defined as the area under the thrust-time curve. From Impulse-Momentum principles in dynamics, the total impulse determines the change in velocity ( $\Delta v$  capability) that a given rocket motor design will provide to a vehicle.

$$I = \int_0^{t_b} F dt = \bar{F} t_b$$

$I$  = Total Impulse (N-s)

$F$  = Instantaneous Thrust Magnitude (N)

$t_b$  = thrust duration or burn time (s)

$\bar{F}$  = Average Thrust Magnitude (N)

“Total Impulse is the Area Under the Thrust Curve”

For any system,

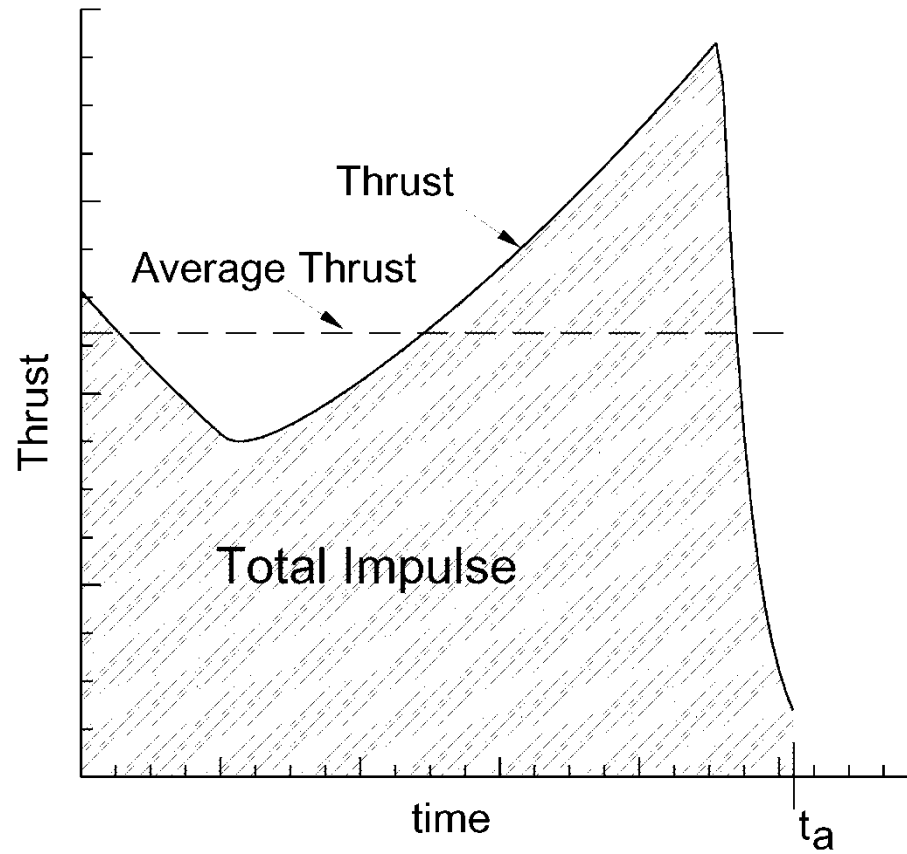
$$I = \bar{F} t_b$$

For a system with a constant thrust magnitude,

$$I = F t_b$$



# Typical Thrust-Time Curve



# Specific Impulse

- The specific impulse is defined to be the thrust divided by the weight flow rate measured at sea level gravity.
- Specific Impulse is defined in this way such that the units are in time (usually seconds) and are therefore insensitive to systems of units.

$$I_{sp} = \frac{\int_0^{t_b} F dt}{g_0 \int_0^{t_b} \dot{m} dt}$$

$F = \text{thrust magnitude (N)}$

$m = \text{propellant mass flow rate (kg/s)}$

$g_0 = 9.807 \text{ (m/s}^2\text{)}$

$$I_{sp} = \frac{T_{avg}}{\dot{m}_{avg} g}$$

For constant thrust and  
constant mass flow rate

$$I_{sp} = \frac{T_{avg}}{\dot{m}_{avg} g} = \frac{\dot{m} c}{\dot{m} g} = \frac{c}{g}$$

| Propulsion                     | $I_{sp}$ (s) |
|--------------------------------|--------------|
| Cold Gas                       | 60-250       |
| Liquid Rocket                  | 140-235      |
| Monopropellant<br>Bipropellant | 320-460      |
| Solid Rocket                   | 260-300      |
| Hybrid Rocket                  | 290-350      |

Typical  $I_{sp}$  values for  
various propulsion  
systems

# Mass Ratio

- The mass ratio is a design performance factor which is defined to be the ratio of the vehicle excluding the propellant divided by the initial total mass of the vehicle.
- This factor along with the specific impulse drive the total vehicle capability as we will see in some examples.

$$MR = \frac{m_f}{m_o} = \frac{\text{final vehicle mass after burnout}}{\text{initial total vehicle mass}}$$

## Note:

$$m_p + m_f = m_o \quad \text{and} \quad \frac{m_p}{m_o} + \frac{m_f}{m_o} = 1 \quad \text{or} \quad \frac{m_p}{m_o} = 1 - \frac{m_f}{m_o} = 1 - MR$$

# Characteristic Velocity

Let's revisit the mass flow rate equation for an isentropic nozzle

$$\frac{\dot{m}}{A} = \frac{p_o \sqrt{\gamma}}{\sqrt{RT_o}} M \left( 1 + \frac{\gamma-1}{2} M^2 \right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

$$\dot{m} = \frac{A^* p_o \sqrt{\gamma}}{\sqrt{RT_o}} \left( \frac{\gamma+1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}} = p_o A^* \left[ \frac{\sqrt{\gamma}}{\sqrt{RT_o}} \left( \frac{\gamma+1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}} \right]$$

Notice that the bracketed term has units of inverse velocity. This term is entirely a function of the propellant properties and the chamber (propellant combustion) temperature and is thus a characteristic property of the propellant. We write this property as:

$$\frac{\sqrt{RT_o}}{\sqrt{\gamma} \left( \frac{\gamma+1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}}} = c^* = \text{characteristic flow velocity}$$

# Characteristic Velocity

A property of the gas can be defined to simplify the gas specific heat ratio function.

$$\Gamma = \sqrt{\gamma} \left( \frac{\gamma+1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

“Cap gamma” is a function of the specific heat ratio only and ranges from about 0.60 to 0.90 and is 0.684 for air and is about 0.65 for most rocket flows.

$c^*$  can then be written as:

$$c^* = \frac{\sqrt{RT_o}}{\Gamma}$$

Or

$$\Gamma = \frac{\sqrt{RT_o}}{c^*}$$

# Characteristic Velocity

- The characteristic velocity can be used to simplify the mass flow rate expression.

$$\dot{m} = \frac{p_o A^*}{c^*}$$

or

$$c^* = \frac{p_o A_t}{\dot{m}}$$

**This will lead to a more succinct expression for thrust in some instances**

# Thrust Coefficient

- One way to describe the principal design variables for a chemical rocket is as follows:
  - Exit pressure
  - Chamber pressure
  - Exit area
  - Gas properties ( $T_o$ ,  $\gamma$ ,  $R(\text{MW})$ ,  $c_p(\text{MW}, \gamma)$ )

Express exit velocity in terms of these design variables using the energy equation:

$$u_e = \sqrt{2c_p(T_o - T_e)} = \sqrt{2c_p T_o \left(1 - \left(\frac{T_e}{T_o}\right)\right)}$$

Assuming  
Isentropic  
Flow

$$u_e = \sqrt{2c_p T_o \left(1 - \left(\frac{p_e}{p_o}\right)^{\frac{\gamma-1}{\gamma}}\right)} = \sqrt{2 \frac{\gamma}{\gamma-1} R T_o \left(1 - \left(\frac{p_e}{p_o}\right)^{\frac{\gamma-1}{\gamma}}\right)}$$

\*

# Thrust Coefficient

Rewrite the mass flow rate for an isentropic nozzle as:

$$\dot{m} = p_o A^* \left[ \frac{\sqrt{\gamma}}{\sqrt{RT_o}} \left( \frac{\gamma+1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}} \right] = \frac{p_o A^*}{\sqrt{RT_o}} \sqrt{\gamma \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{(\gamma-1)}}} \quad ***$$

Substitute \* and \*\* into the thrust equation

$$F_t = \dot{m} u_e + A_e (p_e - p_a)$$

$$F_t = p_o A^* \sqrt{\left( \frac{\gamma}{RT_o} \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{(\gamma-1)}} \right) \left( \frac{2\gamma RT_o}{\gamma-1} \left( 1 - \left( \frac{p_e}{p_o} \right)^{\frac{\gamma-1}{\gamma}} \right) \right)} + A_e (p_e - p_a)$$

$$F_t = p_o A^* \sqrt{\frac{2\gamma^2}{\gamma-1} \left( \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{(\gamma-1)}} \right) \left( 1 - \left( \frac{p_e}{p_o} \right)^{\frac{\gamma-1}{\gamma}} \right)} + A_e (p_e - p_a)$$



# Thrust Coefficient

**The thrust coefficient arises from nondimensionalizing this form of the thrust equation.**

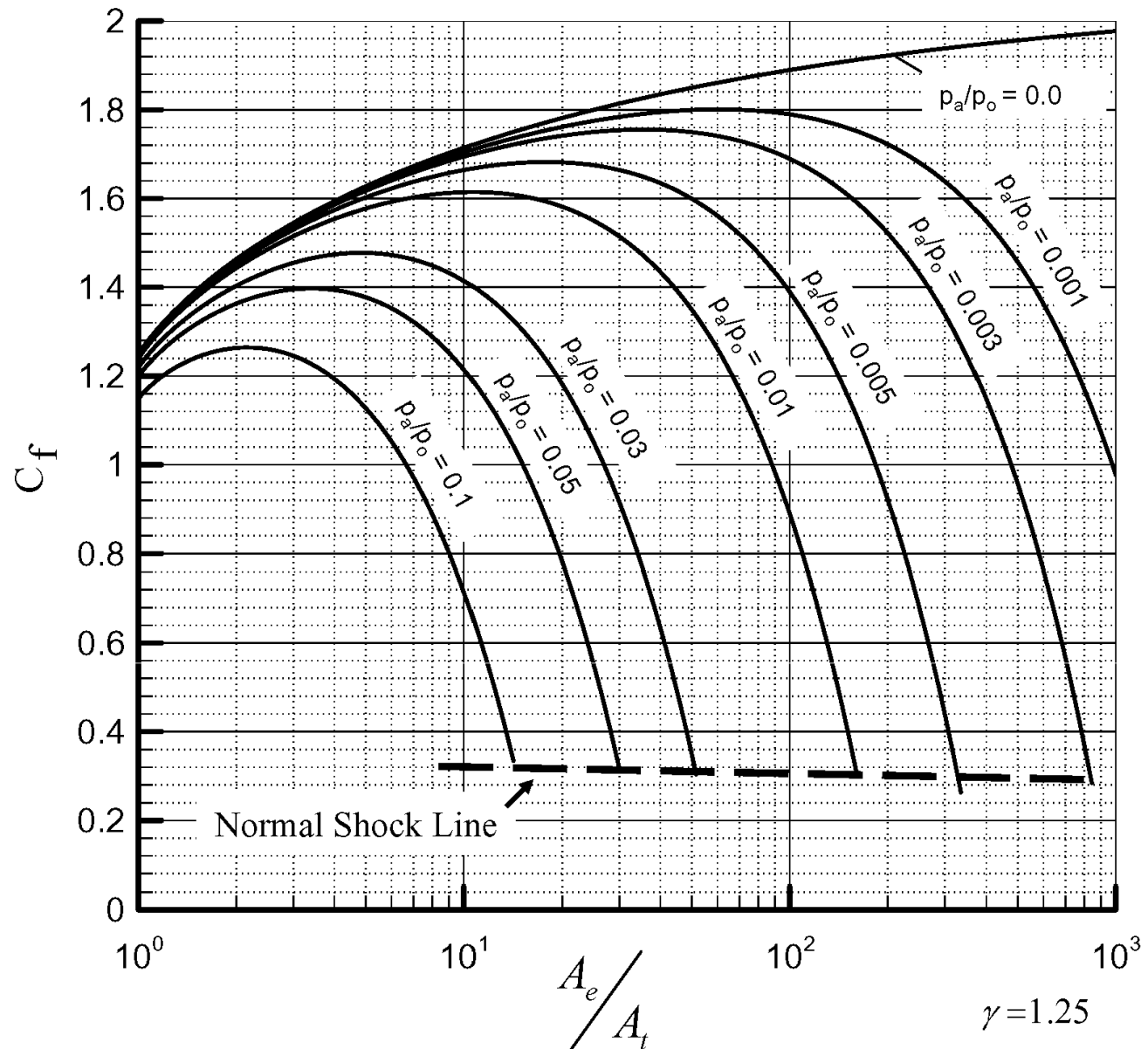
$$\frac{F_t}{p_o A^*} = \sqrt{\frac{2\gamma^2}{\gamma-1} \left( \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \right) \left( 1 - \left( \frac{p_e}{p_o} \right)^{\frac{\gamma-1}{\gamma}} \right)} + \frac{A_e}{A^*} \left( \frac{p_e}{p_o} - \frac{p_a}{p_o} \right) = C_F$$

**The thrust coefficient is a function entirely of the rocket design parameters: propellant (specific heat ratio); pressure ratio; and nozzle expansion ratio.**

**Note that, since**  $\dot{m}c^* = p_o A^*$  **The thrust can be written as:**

$$F_t = p_o A^* C_F = \dot{m}c^* C_F$$

# Thrust Coefficient



# Efficiency

- **Thermal Efficiency-** The source of the energy for chemical propulsion is the stored energy in the propellant mostly contained in the chemical bond energy. This energy is converted to directed kinetic energy in the thrust chamber (combustor and nozzle). The percentage of stored chemical energy which is converted into directed kinetic energy is known as the thermal efficiency or:

$$\eta_{th} = \frac{\text{rate of production of KE}}{\text{rate of consumption of chemical energy}} = \frac{\frac{1}{2} \dot{m} u_e^2}{\dot{m} Q_R} = \frac{u_e^2}{2 Q_R}$$

where  $Q_R$  is the “heat content” of the propellant

# Efficiency

- **Propulsive Efficiency-** The production of kinetic energy does not explicitly produce thrust. Thrust is associated with a momentum exchange which is in turn related to the kinetic energy. The rate at which the kinetic energy is converted into thrust power is referred to as the propulsive efficiency or:

$$\eta_{prop} = \frac{(Thrust)(flight\ velocity)}{KE\ production\ rate} = \frac{Tu}{\frac{1}{2}\dot{m}u_e^2}$$

**For an ideal rocket (nozzle), this becomes:**

$$\eta_{prop} = \frac{\dot{m}u_e u}{\frac{1}{2}\dot{m}u_e^2} = 2 \frac{u}{u_e} = 2 \frac{u}{c}$$

# Efficiency

- **Overall Efficiency-** The overriding consideration is the rate at which chemical energy is converted into thrust power or:

$$\begin{aligned}\eta_{overall} &= \frac{(Thrust)(flight\ velocity)}{rate\ of\ consumption\ of\ chemical\ energy} \\ &= \frac{Tu}{\dot{m}Q_R} = \left( \frac{Tu}{\frac{1}{2}\dot{m}u_e^2} \right) \left( \frac{\frac{1}{2}\dot{m}u_e^2}{\dot{m}Q_R} \right) \\ &= \eta_{prop}\eta_{th}\end{aligned}$$

# Summary of Performance

- The thrust for a chemical rocket can most conveniently be represented in any of several ways depending on available data.

$$F_t = \dot{m}u_e + A_e(p_e - p_a)$$

Fundamental flow parameters

$$F_t = \dot{m}c$$

Effective exhaust velocity

$$F_t = p_o A^* C_F$$

Thrust coefficient and chamber condition

$$F_t = \dot{m}c^* C_F$$

Thrust coefficient and propellant parameters

$$F_{t_{avg}} = \dot{m}_{avg} g I_{sp}$$

Specific Impulse

$$F_{avg} = \frac{I_{total}}{t_b}$$

Total Impulse

$$F_t = \eta_{overall} \left( \frac{\dot{m}Q_R}{u} \right)$$

Efficiency

**Note: Thrust coefficient is not directly dependent on flame temperature but specific impulse is.**

# Summary: Relationships Between Performance Parameters

|                               | <b>A*</b><br>Throat Area          | <b>C</b><br>Effective Exhaust Velocity | <b>C*</b><br>Characteristic Exhaust Velocity | <b>C<sub>F</sub></b><br>Thrust Coefficient | <b>F<sub>t</sub></b><br>Thrust | <b>I<sub>sp</sub></b><br>Specific Impulse | <b>P<sub>c</sub></b><br>Chamber Pressure | <b><math>\dot{m}</math></b><br>Mass Flow Rate |
|-------------------------------|-----------------------------------|----------------------------------------|----------------------------------------------|--------------------------------------------|--------------------------------|-------------------------------------------|------------------------------------------|-----------------------------------------------|
| <b>A* =</b>                   |                                   | $\frac{c\dot{m}}{P_c C_F}$             | $\frac{c^* \dot{m}}{P_c}$                    | $\frac{F_t}{C_F P_c}$                      | $\frac{F_t}{C_F P_c}$          | $\frac{I_{sp} \dot{m} g}{P_c C_F}$        | $\frac{F_t}{C_F P_c}$                    | $\frac{\dot{m} c^*}{P_c}$                     |
| <b>C =</b>                    | $\frac{A^* P_c C_F}{\dot{m}}$     |                                        | $c^* C_F$                                    | $c^* C_F$                                  | $\frac{F_t}{\dot{m}}$          | $I_{sp} g_o$                              | $\frac{A^* P_c C_F}{\dot{m}}$            | $\frac{F_t}{\dot{m}}$                         |
| <b>C* =</b>                   | $\frac{A^* P_c}{\dot{m}}$         | $\frac{c}{C_F}$                        |                                              | $\frac{c}{C_F}$                            | $\frac{F_t}{C_F \dot{m}}$      | $\frac{I_{sp} g_o}{C_F}$                  | $\frac{A^* P_c}{\dot{m}}$                | $\frac{F_t}{\dot{m} C_F}$                     |
| <b>C<sub>F</sub> =</b>        | $\frac{F_t}{A^* P_c}$             | $\frac{c}{c^*}$                        | $\frac{c}{c^*}$                              |                                            | $\frac{F_t}{A^* P_c}$          | $\frac{I_{sp} g_o}{c^*}$                  | $\frac{F_t}{A^* P_c}$                    | $\frac{F_t}{\dot{m} c^*}$                     |
| <b>F<sub>t</sub> =</b>        | $A^* P_c C_F$                     | $\dot{m} c$                            | $\dot{m} c^* C_F$                            | $A^* P_c C_F$                              |                                | $I_{sp} \dot{m} g_o$                      | $A^* P_c C_F$                            | $I_{sp} \dot{m} g_o$                          |
| <b>I<sub>sp</sub> =</b>       | $\frac{A^* P_c C_F}{\dot{m} g_o}$ | $\frac{c}{g_o}$                        | $\frac{c^* C_F}{g_o}$                        | $\frac{c^* C_F}{g_o}$                      | $\frac{F_t}{\dot{m} g_o}$      |                                           | $\frac{A^* P_c C_F}{\dot{m} g_o}$        | $\frac{F_t}{\dot{m} g_o}$                     |
| <b>P<sub>c</sub> =</b>        | $\frac{F_t}{A^* C_F}$             | $\frac{c\dot{m}}{C_F A^*}$             | $\frac{c^* \dot{m}}{A^*}$                    | $\frac{F_t}{C_F A^*}$                      | $\frac{F_t}{C_F A^*}$          | $\frac{I_{sp} \dot{m} g_o}{C_F A^*}$      |                                          | $\frac{c^* \dot{m}}{A^*}$                     |
| <b><math>\dot{m}</math> =</b> | $\frac{A^* P_c}{c^*}$             | $\frac{F_t}{c}$                        | $\frac{F_t}{C_F c^*}$                        | $\frac{F_t}{C_F c^*}$                      | $\frac{F_t}{I_{sp} g_o}$       | $\frac{F_t}{I_{sp} g_o}$                  | $\frac{A^* P_c}{c^*}$                    |                                               |

## Performance Example 3

The following data are given for a certain rocket unit: thrust = 8896 kN: propellant consumption = 3867 kg/s: velocity of vehicle = 400 m/s: energy content of propellant = 6.911 MJ/kg. Determine:

- a) the effective exhaust velocity
- b) the kinetic jet energy rate per unit flow of propellant
- c) the thermal efficiency
- d) the propulsive efficiency
- e) the overall efficiency
- f) the specific impulse.



## Solution

a) Effective Exhaust velocity can be written as:

$$c = \frac{F_t}{\dot{m}} = \frac{8,896,000 \text{ N}}{3,867 \text{ kg/s}} = 2300.5 \text{ m/s}$$

b) kinetic jet energy rate per unit flow rate

$$\frac{\frac{1}{2} \dot{m} u_e^2}{\dot{m}} = \frac{1}{2} u_e^2 = \frac{1}{2} c^2 = \frac{1}{2} (2300.5 \text{ m/s})^2 = 2646 \text{ kJ/kg}$$

c) Thermal Efficiency

$$\eta_{th} = \frac{\frac{u_e^2}{2}}{Q_R} = \frac{2.646 \text{ MJ/kg}}{6.911 \text{ MJ/kg}} = 0.383$$

## Solution

### d) propulsive efficiency

$$\eta_{prop} = \frac{Tu}{\frac{1}{2}\dot{m}u_e^2} = \frac{(8,896,000\text{ N})(400\text{ m/s})}{\frac{1}{2}(3867\text{ kg/s})(2300.5\text{ m/s})} = 0.348$$

### e) Overall efficiency

$$\eta_{overall} = \eta_{prop}\eta_{th} = 0.133$$

### f) Specific Impulse

$$I_{sp} = \frac{T_{avg}}{\dot{m}_{avg}g} = \frac{c}{g} = \frac{2300.5\text{ m/s}}{9.81\text{ m/s}^2} = 234.5\text{ s}$$

## Performance Example 4

For an ideal rocket with a characteristic velocity of 1500 m/s, a nozzle throat diameter of 18 cm, a thrust coefficient of 1.38, and a mass flow rate of 40 kg/s, determine the chamber pressure, thrust and specific impulse.

**Solution:**

**Chamber Pressure:**

$$\dot{m} = \frac{p_o A^*}{c^*} \quad \text{or} \quad p_o = \frac{\dot{m} c^*}{A^*} = \frac{\dot{m} c^*}{\left( \frac{\pi d_t^2}{4} \right)} = \frac{(40 \text{ kg/s})(1500 \text{ m/s})}{\left( \frac{\pi (0.18 \text{ m})^2}{4} \right)} = 2,357 \text{ kPa}$$

## Performance Example 4 Solution

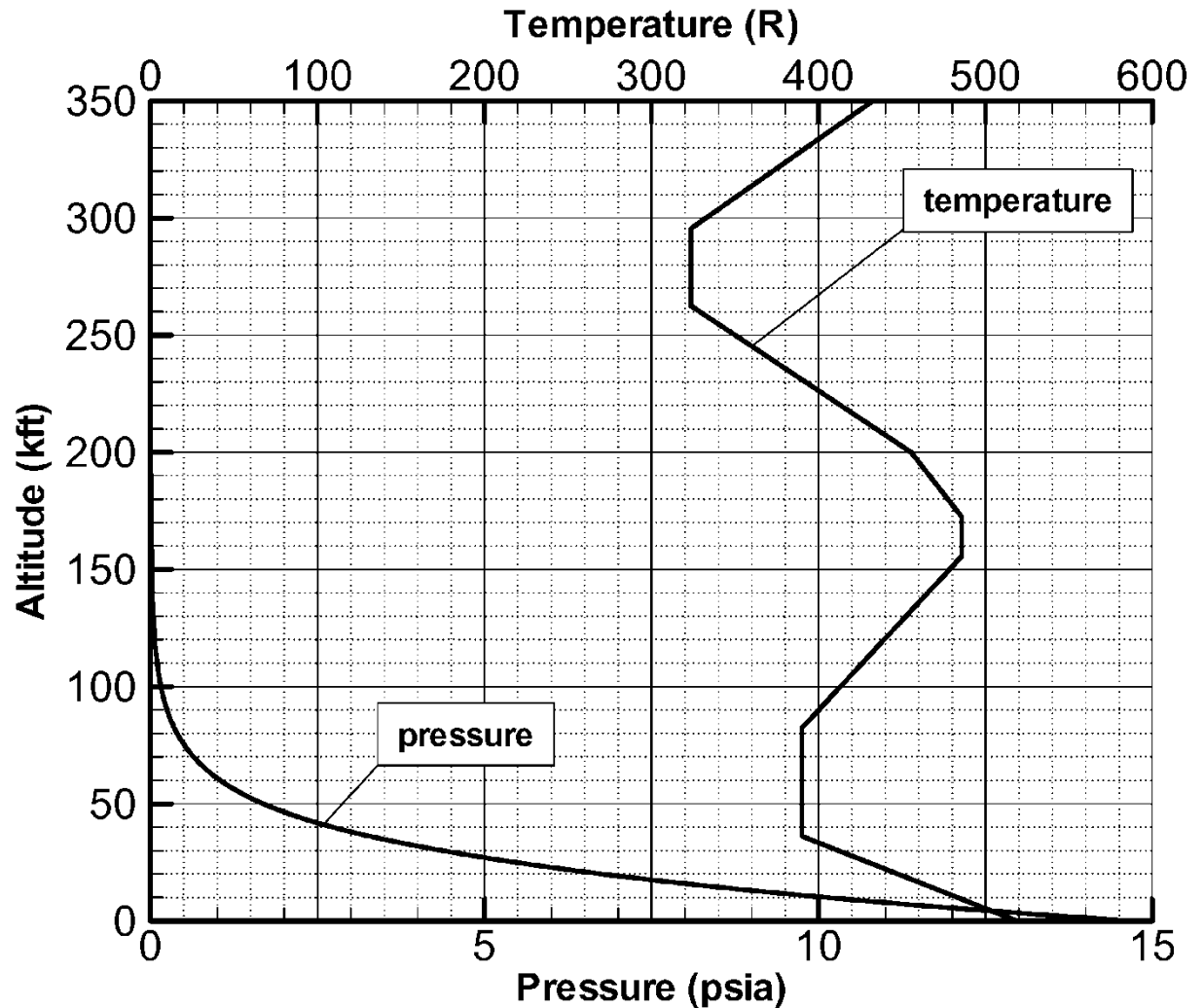
$$F_t = p_o A^* C_F = (2,357 \text{ kPa}) \left( \frac{\pi (0.18 \text{ m})^2}{4} \right) (1.38) = 82.6 \text{ kN}$$

### Specific Impulse

$$I_{sp} = \frac{T}{\dot{m}g} = \frac{82,600 \text{ N}}{(40 \text{ kg/s})(9.81 \text{ m/s}^2)} = 210.5 \text{ s}$$

# Standard Atmosphere

Rocket thrust is dependent on altitude through the ambient pressure term. It is thus important to define an atmospheric model, such as the “standard atmosphere” shown at right.



# Potential Chemical Improvements

- **Examination of Specific Impulse:**
  - The basic performance measure is the specific impulse.
  - One of the ways to represent specific impulse is:

$$I_{sp} = \frac{c^* C_T}{g_e} = \frac{c}{g_e} \approx \frac{u_e}{g_e} \text{ (high nozzle pressure ratio)}$$

# Potential Chemical Improvements

- Specific Impulse can be related more directly to propellant properties by looking at  $u_e$ .

$$u_e = \sqrt{\frac{2\gamma \frac{R_u}{MW} T_o}{\gamma - 1} \cdot \left[ 1 - \left( \frac{p_e}{p_o} \right)^{\frac{\gamma-1}{\gamma}} \right]}$$

$$(u_e)_{\max} = \sqrt{\frac{2\gamma \frac{R_u}{MW} T_o}{\gamma - 1}}$$

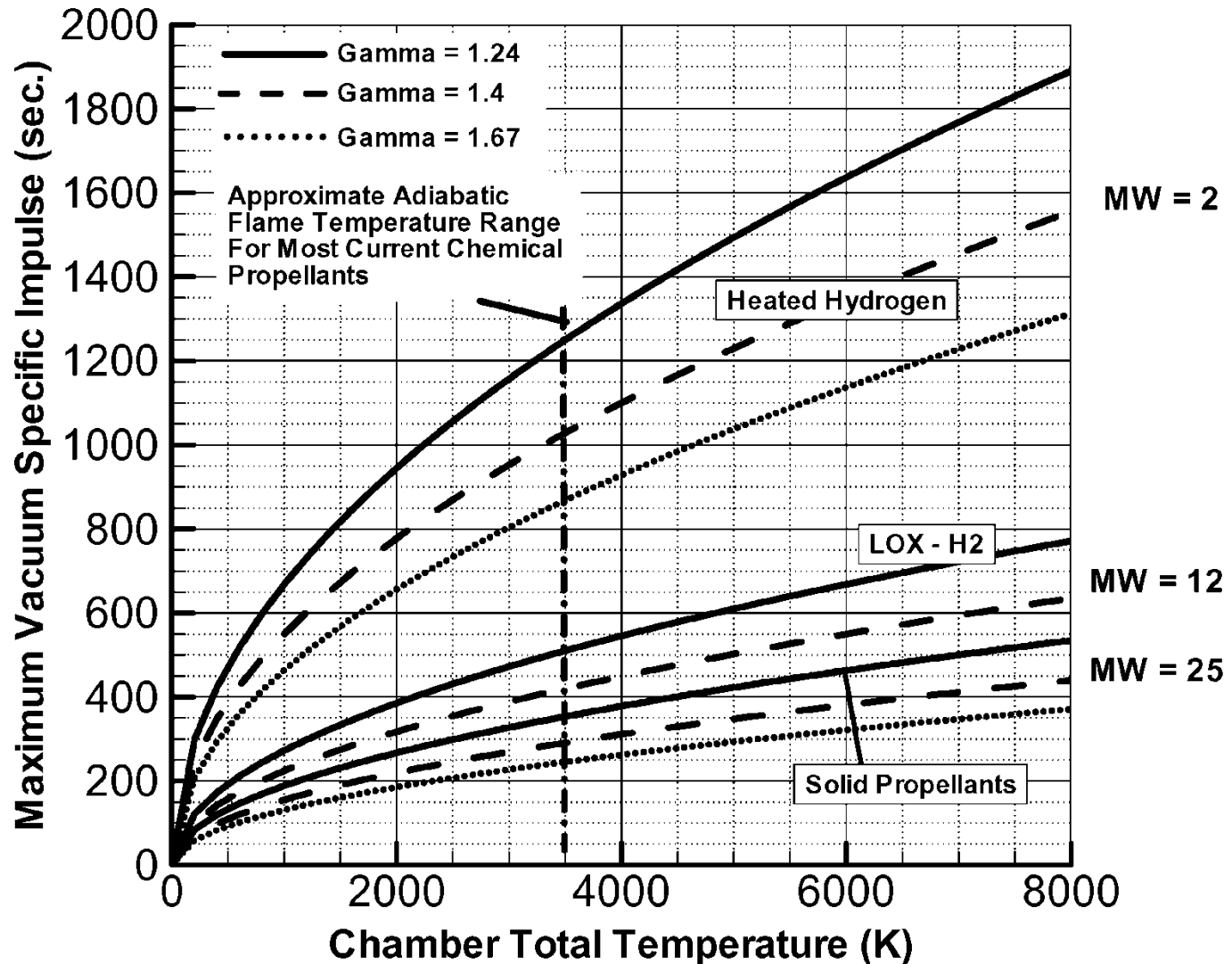
# Chemical Improvements

- Clearly, in the limiting case of high chamber pressure, the specific impulse is a function of propellant properties only:
  - I.e.:

$$I_{sp} = f(\gamma, MW, T_o)$$



# Specific Impulse of an Ideal Rocket Motor as a Function of Propellant Properties



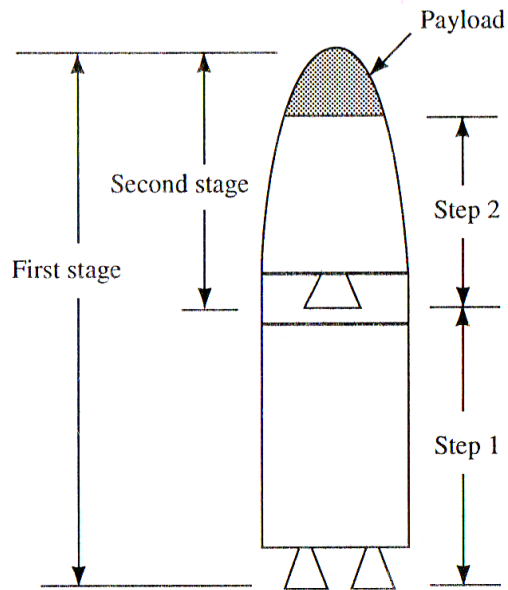
# Staging

- A stage is defined as all of the vehicle that is being accelerated by the rocket engines.
- A step is the stage less the payload for that stage.
- Steps are jettisoned through the process commonly thought of as staging.
- When describing the physical characteristics, the term stage is generally used to refer to the physical characteristics of the stage.



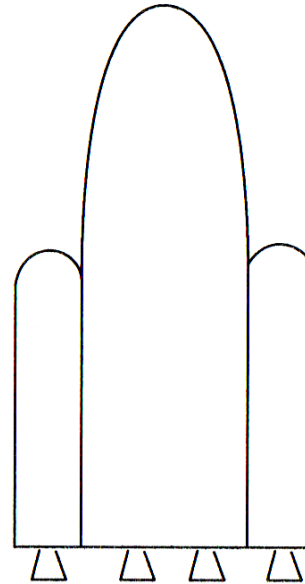
# Types of Staging

## Tandem



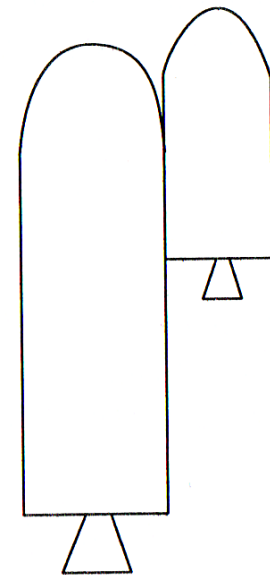
Hale p. 169

## Parallel



Hale p. 171

## Piggyback



# Staging

The basic advantage of multiple staging for delivery of a specified payload to a specified state is that a percentage of the total structure can be dropped off along the way. That is, not all of the structure need be accelerated to the payload delivery condition. To develop an analysis of the advantage of staging, we review and define some single stage parameters:

$$MR = \frac{m_f}{m_o} = \frac{m_s + m_L}{m_s + m_L + m_p} = \frac{\text{final vehicle mass after burnout}}{\text{initial total vehicle mass}}$$

$$\frac{m_{prop}}{m_o} = 1 - MR = 1 - e^{-v_f/u_e} \quad (9)$$

$$MR = e^{-v_f/u_e} \quad (10)$$

# Staging

**For a single stage:**

$$MR = e^{-v_f/u_e} \Rightarrow \Delta v_{stage} = -(u_e)_{stage} \ln(MR_{stage})$$

$$\Delta v_{stage} = -(u_e)_{stage} \ln(MR_{stage})$$

**For all stages:**

$$\Delta v_T = \sum \Delta v_{stage} = \sum_{j=1}^n -(u_e)_j \ln(MR_j)$$

**Where j is the j<sup>th</sup> stage and n is the number of stages**

**In the case where all stage specific impulses are the same and we have assumed  $u_e = c$ ,**

$$\begin{aligned} -\frac{\Delta v_T}{u_e} &= -\frac{\Delta v_T}{c} = \sum_{j=1}^n \ln(MR_j) = \ln(MR_1) + \ln(MR_2) + \dots + \ln(MR_n) \\ &= \ln(MR_1 \times MR_2 \times \dots \times MR_n) \end{aligned}$$

# Stage Terminology

- **Restricted Staging** refers to the use of stages which have identical propulsive characteristics and mass ratios.
- **Generalized Staging** refers to staging using variable mass ratios and propellant/performance characteristics

# Staging

Definition of payload ratio and structural ratio  $\lambda_i$  and  $\varepsilon_i$

**Payload ratio, single stage**

$$\lambda = \frac{m_L}{m_o - m_L} = \frac{m_L}{m_p + m_s}$$

**Structure ratio, single stage**

$$\varepsilon = \frac{m_s}{m_o - m_L} = \frac{m_{burnout} - m_L}{m_p + m_s}$$

**Payload ratio,  $j^{\text{th}}$  stage**

$$\lambda_j = \frac{m_{o(j+1)}}{m_{oi} - m_{o(j+1)}}$$

**Structure ratio,  $j^{\text{th}}$  stage**

$$\varepsilon_i = \frac{m_{si}}{m_{oi} - m_{o(i+1)}}$$

# Multi-stage Example Problem, H&P

It can be shown that the optimum arrangement for multi-staging is that the payload and structural ratio  $\lambda_i$  and  $\varepsilon_i$  be the same for all stages. Consider the effect of a departure from this optimum arrangement for a three stage rocket with the same engine specific impulse (400 sec) for all three stages. The mass distribution for the stages is as follows:

| Stage      | 1       | 2      | 3     |
|------------|---------|--------|-------|
| $M_o$ (kg) | 100,000 | 18,000 | 4,000 |
| $M_p$ (kg) | 75,000  | 12,000 | 2,000 |
| $M_L$ (kg) |         |        | 1,025 |

- For each stage determine  $\lambda_i$  and estimate the total velocity increment for the three stages neglecting drag and gravity effects.
- Compare this velocity increment to that of an optimized three stage vehicle with the same specific impulse, payload mass, total propellant mass and total structural mass.



# Multi-stage Example Problem, Solution

|                           |                                                      | Stage 1       | Stage 2      | Stage 3      |
|---------------------------|------------------------------------------------------|---------------|--------------|--------------|
| <b>Payload ratio</b>      | $\lambda_j = \frac{m_{o(i+1)}}{m_{oi} - m_{o(i+1)}}$ | <b>0.220</b>  | <b>0.286</b> | <b>0.345</b> |
| <b>Structure ratio</b>    | $\varepsilon_i = \frac{m_{si}}{m_{oi} - m_{o(i+1)}}$ | <b>0.0854</b> | <b>0.143</b> | <b>0.327</b> |
| <b>Velocity increment</b> | $\Delta v_{st} = -(u_e)_{st} \ln(MR_{st})$           | <b>5434</b>   | <b>4308</b>  | <b>2722</b>  |

$$\Delta v_T = \sum \Delta v_{stage} = 5434 + 4308 + 2722 = 12,460 \text{ m/s}$$

# Multi-stage Example Problem, Solution

## Optimized Vehicle

| Mass  | Stage1 | Stage 2 | Stage 3 | Total   |
|-------|--------|---------|---------|---------|
| $m_p$ | 75,000 | 12,000  | 2,000   | 89,000  |
| $m_s$ | 7,000  | 2,000   | 975     | 9,975   |
| $m_L$ |        |         |         | 1,025   |
| $m_o$ |        |         |         | 100,000 |

**Note:**

$$\lambda_j = \frac{m_{o(j+1)}}{m_{oi} - m_{o(j+1)}} = \frac{1}{\frac{m_{oi}}{m_{o(j+1)}} - 1} \Rightarrow \frac{m_{oi}}{m_{o(j+1)}} = \frac{1}{\lambda_j} + 1 = \frac{1 + \lambda_j}{\lambda_j}$$

**For an optimized vehicle,  $\lambda_j$  is constant and we can write:**

$$\frac{m_o}{m_L} = \left( \frac{m_{o1}}{m_{o(2)}} \right) \left( \frac{m_{o2}}{m_{o(3)}} \right) \left( \frac{m_{o3}}{m_L} \right) = \left( \frac{1 + \lambda_j}{\lambda_j} \right)^3 = \frac{100,000}{1,025} \Rightarrow \lambda = 0.2775$$

# Multi-stage Example Problem, Solution

**For a single stage**

$$\lambda = \frac{m_L}{m_o - m_L} = \frac{1}{\frac{m_o}{m_L} - 1} \Rightarrow \frac{m_o}{m_L} = \frac{1 + \lambda}{\lambda}$$

**For multiple stages:**

$$\frac{m_{o1}}{m_L} = \left( \frac{m_{o1}}{m_{o2}} \right) \left( \frac{m_{o2}}{m_{o3}} \right) \left( \frac{m_{o3}}{m_L} \right) = \left( \frac{1 + \lambda}{\lambda} \right)^3 = \frac{100,000}{1,025} \Rightarrow \lambda = 0.2775$$

**Structure ratio:**

$$\varepsilon = \frac{m_{s1}}{m_{p1} + m_{s1}} = \frac{m_{s2}}{m_{p2} + m_{s2}} = \frac{m_{s3}}{m_{p3} + m_{s3}}$$

**Rearranging:**

$$m_{p1} = m_{s1} \left( \frac{1 - \varepsilon}{\varepsilon} \right) \quad m_{p2} = m_{s2} \left( \frac{1 - \varepsilon}{\varepsilon} \right) \quad m_{p3} = m_{s3} \left( \frac{1 - \varepsilon}{\varepsilon} \right)$$

**For multiple stages:**

$$m_p = \sum m_{pi} = \left( \frac{1 - \varepsilon}{\varepsilon} \right) \sum m_{si} = m_s \left( \frac{1 - \varepsilon}{\varepsilon} \right)$$

**Solving for  $\varepsilon$ :**

$$\frac{m_p}{m_s} = \left( \frac{1 - \varepsilon}{\varepsilon} \right) = \frac{89,000}{9,975} \Rightarrow \varepsilon = 0.1008$$

# Multi-stage Example Problem, Solution

Solving for the final velocity can be done by looking at a single stage

$$\begin{aligned}\Delta v_T &= \sum_{j=1}^n - (u_e)_j \ln(MR_j) = - (u_e) (\ln(MR_1) + \ln(MR_2) + \ln(MR_3)) \\ &= - (u_e) \ln(MR)^3 = -3(u_e) \ln(MR) = -3(u_e) \ln\left(\frac{m_s + m_L}{m_s + m_L + m_p}\right) \\ &= 3(u_e) \ln\left(\frac{m_s + m_p + m_L}{m_s + m_L}\right) = 3(u_e) \ln\left(\frac{\frac{m_s + m_p}{m_s + m_p} + \frac{m_L}{m_s + m_p}}{\frac{m_s}{m_s + m_p} + \frac{m_L}{m_s + m_p}}\right) \\ &= 3g(I_{sp}) \ln\left(\frac{1 + \lambda}{\varepsilon + \lambda}\right) = 3(9.81)(400) \ln\left(\frac{1 + 0.2775}{0.1008 + 0.2775}\right) = 14,330 \text{ m/s}\end{aligned}$$

**This is a 15% increase over the unoptimized example!**

# Optimization Multiple Stage Rockets

## (After Hill and Peterson)

### Assumptions:

1. Same exhaust velocity for all stages
2. Same structural coefficient for all stages

Velocity increment  
for a given stage:

$$\Delta v_i = u_e \ln(MR) = u_e \ln\left(\frac{1 + \lambda}{\varepsilon + \lambda}\right) \quad (i)$$

(See previous example for this development)

Final velocity after  
the  $n^{\text{th}}$  stage:

$$\frac{v_n}{u_e} = \sum_1^n \ln\left(\frac{1 + \lambda_i}{\varepsilon + \lambda_i}\right) = \sum_1^n F(\lambda_i) \quad (ii)$$

Now employ calculus of variations to  
find the optimum payload ratios,  
(to maximize  $v_n$ )  $\lambda_i$  subject to the  
constraint:

$$\frac{m_L}{m_o} = \prod_1^n \left(\frac{\lambda_i}{1 + \lambda_i}\right)$$

$$\ln\left(\frac{m_L}{m_o}\right) = \sum_1^n \ln\left(\frac{\lambda_i}{1 + \lambda_i}\right) = \sum_1^n G(\lambda_i) \quad (iii)$$

# Optimization Multiple Stage Rockets

In practice, we are maximizing the function  $\sum_1^n F(\lambda_i)$  subject to the constraint in equation (iii) by using the method of Lagrange multipliers which is formed as follows:

$$L(\lambda_i) = F(\lambda_i) + \alpha G(\lambda_i) \quad (iv)$$

( $\alpha$  is the Lagrange multiplier)

For maximum  $v_n$

$$\frac{\partial L}{\partial \lambda_i} = \frac{\partial F}{\partial \lambda_i} + \alpha \frac{\partial G}{\partial \lambda_i} = 0$$

Substituting from eqs. (ii) and (iii):

$$\frac{1}{1 + \lambda_i} - \frac{1}{\varepsilon + \lambda_i} + \frac{\alpha}{\lambda_i} - \frac{\alpha}{1 + \lambda_i} = 0 \quad (v)$$

or

$$\lambda_i = \frac{\alpha \varepsilon}{1 - \alpha - \varepsilon} \quad (vi)$$

# Optimization Multiple Stage Rockets

Since all of the terms in equation (vi) are constant,  $\lambda_i$  must be constant. Setting  $\lambda_i = \lambda$  produces:

$$\frac{m_L}{m_o} = \prod_1^n \left( \frac{\lambda}{1 + \lambda} \right) = \left( \frac{\lambda}{1 + \lambda} \right)^n$$

or

$$\lambda = \frac{\left( \frac{m_L}{m_{o1}} \right)^{\frac{1}{n}}}{1 - \left( \frac{m_L}{m_{o1}} \right)^{\frac{1}{n}}}$$